

Final Network Project Report

Cisco Packet Tracer — Multi-Site Enterprise Design

Student Information

Name: *Yusuf*

GitHub: github.com/engYSF

Contents

1	Executive Summary	2
2	Network Topology	2
2.1	Logical Overview	2
2.2	Site Descriptions	3
3	VLAN Segmentation & IP Addressing	3
3.1	VLAN Table	3
3.2	WAN & Interconnect Links	3
4	Router Configurations	3
4.1	R-MAIN — Router-on-a-Stick + EIGRP	3
4.2	R-EDGE — EIGRP↔RIP Redistribution + PAT	4
4.3	R-BRANCH — VLAN 50 + RIPv2	5
4.4	R-INTERNET (ISP) — Return Route	6
5	Testing & Verification	6
5.1	Routing Table Verification	6
5.2	NAT Translation Table	6
5.3	DHCP Pool Verification	7
5.4	End-to-End Reachability	7
6	Challenges & Resolutions	7
7	Key Lessons Learned	8
	Copyright & Attribution	8

1 Executive Summary

Project Overview

This report documents a **two-site enterprise network** built and verified in Cisco Packet Tracer. The design achieves clean VLAN segmentation, predictable multi-protocol routing, secure Internet breakout, and centralized services — all validated through CLI output and end-to-end reachability tests.

The network is composed of three routing domains interconnected through a central Edge router:

- **Main Office (R-MAIN)** — router-on-a-stick providing inter-VLAN routing for six VLANs; runs **EIGRP AS 10**.
- **Branch (R-BRANCH)** — single-VLAN site running **RIPv2**.
- **Edge Router (R-EDGE)** — redistributes routes between EIGRP and RIP; provides Internet access via **PAT** (overload NAT) to the ISP.
- **ISP / R-INTERNET** — simulates the public Internet; holds a static summary return route (**192.168.0.0/16**) and a loopback **8.8.8.8** (DNS target).
- **Centralized Services** — DHCP, DNS, and HTTP hosted at **192.168.100.10** (Servers VLAN); reachable via `ip helper-address` on all sub-interfaces.

2 Network Topology

2.1 Logical Overview

The diagram below shows the complete logical topology as simulated in Cisco Packet Tracer.

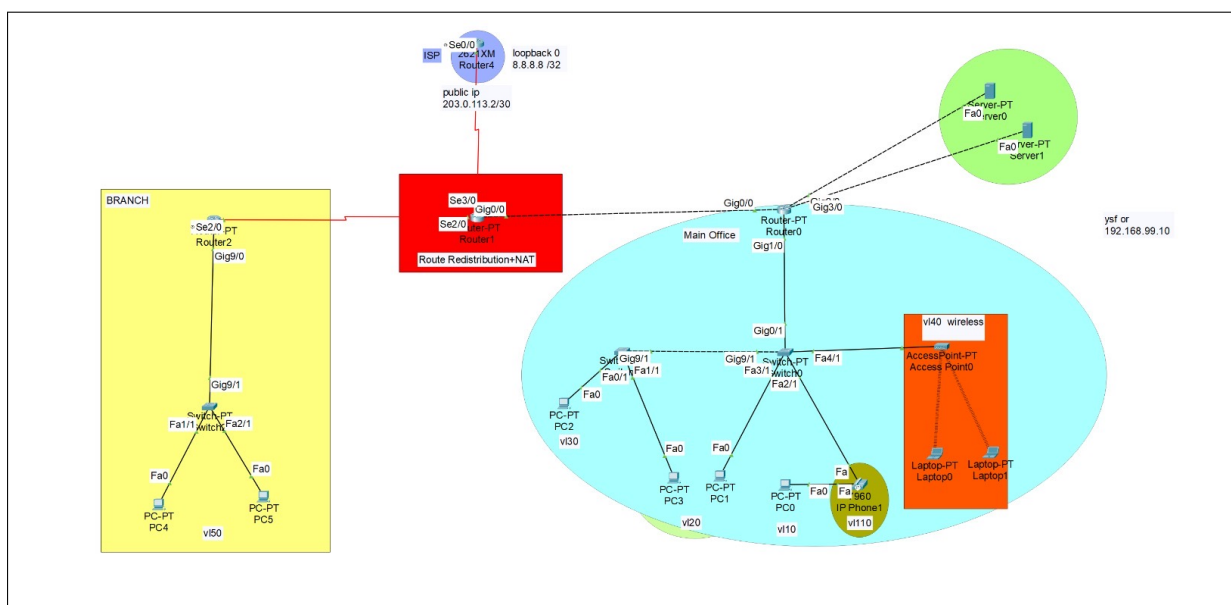


Figure 1: Full Logical Topology — Cisco Packet Tracer

2.2 Site Descriptions

Site / Device	Description
Main Office	Two multilayer switches (Switch0/Switch1), one router (R-MAIN), access point (VLAN 40 wireless), IP Phone (VLAN 110), and server farm (VLAN 100).
Branch	One router (R-BRANCH), one switch, two PCs (PC4, PC5) in VLAN 50.
Edge	Router-only; connects Main Office (GigE), Branch (Serial), and ISP (Serial).
ISP	Single router simulating the Internet with loopback 8.8.8.8/32 .

3 VLAN Segmentation & IP Addressing

3.1 VLAN Table

VLAN ID	Name	Subnet
10	HR	192.168.10.0/24
20	Finance	192.168.20.0/24
30	IT	192.168.30.0/24
40	Wireless	192.168.40.0/24
50	Branch Data	192.168.50.0/24
100	Servers	192.168.100.0/24
110	IP Phone (Voice)	192.168.110.0/24

3.2 WAN & Interconnect Links

Link	Protocol	Subnet
R-MAIN ↔ R-EDGE	GigabitEthernet	10.1.1.0/30
R-EDGE ↔ R-BRANCH	Serial (DCE)	172.16.1.0/30
R-EDGE ↔ ISP	Serial	203.0.113.0/30
ISP Loopback	Loopback0	8.8.8.8/32

4 Router Configurations

4.1 R-MAIN — Router-on-a-Stick + EIGRP

R-MAIN uses sub-interfaces on `Gi1/0` to provide inter-VLAN routing for all local VLANs. EIGRP AS10 advertises all [192.168.x.0/24](#) prefixes and the [10.1.1.0/30](#) WAN link. DHCP relay (`ip helper-address`) forwards client broadcasts to the central server.

```

1 en
2 conf t
3 host R0-MAIN

```

```
4 !
5 int gi1/0.10
6   encap dot1q 10
7   ip add 192.168.10.1 255.255.255.0
8   ip helper-address 192.168.100.10
9 !
10 int gi1/0.20
11   encap dot1q 20
12   ip add 192.168.20.1 255.255.255.0
13   ip helper-address 192.168.100.10
14 !
15 int gi1/0.30
16   encap dot1q 30
17   ip add 192.168.30.1 255.255.255.0
18   ip helper-address 192.168.100.10
19 !
20 int gi1/0.40
21   encap dot1q 40
22   ip add 192.168.40.1 255.255.255.0
23   ip helper-address 192.168.100.10
24 !
25 int gi1/0.100
26   encap dot1q 100
27   ip add 192.168.100.1 255.255.255.0
28 !
29 int gi1/0.110
30   encap dot1q 110
31   ip add 192.168.110.1 255.255.255.0
32   ip helper-address 192.168.100.10
33 !
34 int gi0/0
35   desc To EDGE
36   ip add 10.1.1.1 255.255.255.252
37 !
38 router eigrp 10
39   net 10.1.1.0 0.0.0.3
40   net 192.168.10.0 0.0.0.255
41   net 192.168.20.0 0.0.0.255
42   net 192.168.30.0 0.0.0.255
43   net 192.168.40.0 0.0.0.255
44   net 192.168.100.0 0.0.0.255
45   net 192.168.110.0 0.0.0.255
46   no auto-summary
47 end
48 wr mem
```

4.2 R-EDGE — EIGRP↔RIP Redistribution + PAT

The Edge router is the network's policy hub: it translates between two routing protocols and provides **NAT overload (PAT)** for Internet access. Access-list 1 defines the inside address space; Serial3/0 is the NAT outside interface.

```
1 en
2 conf t
3 host R1-EDGE
4 !
5 int gi0/0
6   desc To R-MAIN
```

```
7  ip add 10.1.1.2 255.255.255.252
8  ip nat inside
9  !
10 int s2/0
11  desc To R-BRANCH
12  ip add 172.16.1.1 255.255.255.252
13  ip nat inside
14  !
15 int s3/0
16  desc To ISP
17  ip add 203.0.113.1 255.255.255.252
18  ip nat outside
19  !
20 router eigrp 10
21  net 10.1.1.0 0.0.0.3
22  redistribute rip metric 10000 100 255 1 1500
23  no auto-summary
24  !
25 router rip
26  ver 2
27  no auto-summary
28  net 172.16.1.0
29  redistribute eigrp 10 metric 2
30  !
31 ip route 0.0.0.0 0.0.0.0 s3/0
32 access-list 1 permit 192.168.0.0 0.0.255.255
33 ip nat inside source list 1 int s3/0 overload
34 end
35 wr mem
```

4.3 R-BRANCH — VLAN 50 + RIPv2

The Branch router terminates a serial WAN link (DCE side: clock rate 64000) and advertises VLAN 50 via RIPv2. The `ip helper-address` forwards DHCP requests to the centralized server at [192.168.100.10](#).

```
1  en
2  conf t
3  host R2-BRANCH
4  !
5  int gi9/0.50
6  encaps dot1q 50
7  ip add 192.168.50.1 255.255.255.0
8  ip helper-address 192.168.100.10
9  !
10 int s2/0
11  desc To EDGE
12  ip add 172.16.1.2 255.255.255.252
13  clock rate 64000
14  !
15 router rip
16  ver 2
17  no auto-summary
18  net 172.16.1.0
19  net 192.168.50.0
20 end
21 wr mem
```

4.4 R-INTERNET (ISP) — Return Route

The ISP router holds a static summary route pointing back through the Edge router, ensuring all [192.168.0.0/16](#) reply traffic is correctly forwarded. The loopback [8.8.8.8](#) simulates a public DNS/ping target.

```
1  en
2  conf t
3  host R-INTERNET
4  !
5  int s0/0
6  ip add 203.0.113.2 255.255.255.252
7  no shut
8  !
9  int lo0
10 ip add 8.8.8.8 255.255.255.255
11 !
12 ip route 192.168.0.0 255.255.0.0 203.0.113.1
13 end
14 wr mem
```

5 Testing & Verification

All test cases below were executed in Cisco Packet Tracer. Expected outcomes matched observed CLI output.

5.1 Routing Table Verification

R-MAIN — Shows all connected VLANs ([192.168.10{110/24}](#)), the WAN to Edge ([10.1.1.0/30](#)), and a D*EX 0.0.0.0/0 default route learned via EIGRP external from Edge (via [10.1.1.2](#)).

R-EDGE — Shows EIGRP internal routes to all HQ VLANs, one RIP route to [192.168.50.0/24](#) (Branch), and a static default S* 0.0.0.0/0 via [203.0.113.2](#).

R-BRANCH — Shows all HQ VLANs via RIP ([120/1](#) via [172.16.1.1](#)) and a R* 0.0.0.0/0 default route.

ISP — Shows S 192.168.0.0/16 via [203.0.113.1](#) and loopback [8.8.8.8](#).

5.2 NAT Translation Table

The Edge router confirms active PAT entries. The output of `show ip nat translations` shows inside local addresses ([192.168.20.x](#), [192.168.30.x](#)) being translated to the single outside global address [203.0.113.1](#) with unique port numbers.

`show ip nat statistics` confirms: 9 dynamic extended translations, inside interfaces Gi0/0 and Se2/0, outside interface Se3/0.

5.3 DHCP Pool Verification

Pool Name	Default GW	DNS Server	Start IP
BRANCH VL50	192.168.50.1	192.168.100.10	192.168.50.50
IP PHONE VL110	192.168.110.1	192.168.100.10	192.168.110.50
Wireless VL40	192.168.40.1	192.168.100.10	192.168.40.50
IT VL30	192.168.30.1	192.168.100.10	192.168.30.50
FIN VL20	192.168.20.1	192.168.100.10	192.168.20.50
HR VL10	192.168.10.1	192.168.100.10	192.168.10.50

5.4 End-to-End Reachability

- **Web by IP:** PC4 (Branch) browses `http://192.168.100.10` — page loads showing “*Final Project — Yusuf*”. [OK]
- **Web by DNS:** PC0 browses `http://ysf` — DNS resolves, page loads. [OK]
- **Traceroute — Branch to VLAN 40:** `tracert 192.168.40.51` from PC4 shows 4 hops: `192.168.50.1 → 172.16.1.1 → 10.1.1.1 → 192.168.40.51`. [OK]
- **Internet (ISP ping):** Clients reach `8.8.8.8` via PAT. [OK]

6 Challenges & Resolutions

The following issues were encountered during implementation and resolved systematically.

Problem	Resolution
Internet replies dropped	Added static summary route on ISP: <code>ip route 192.168.0.0 255.255.0.0 203.0.113.1</code>
Branch couldn't reach HQ	Enabled bidirectional redistribution on Edge (EIGRP ↔ RIP with explicit metrics)
DHCP not working on a VLAN	Added correct <code>ip helper-address 192.168.100.10</code> on the missing sub-interface
NAT not translating	Corrected <code>ip nat inside / ip nat outside</code> assignments and verified ACL 1
STP instability (occasional)	Tightened trunk allowed VLANs; enabled PortFast + BPDU Guard on edge ports

7 Key Lessons Learned

1. **Route redistribution requires metric seeding.** When moving routes between EIGRP and RIP, both directions must include explicit metrics; missing a metric causes the redistributed routes to be silently dropped.
2. **NAT interface roles are critical.** Assigning `ip nat inside / outside` to the wrong interfaces is the most common NAT failure mode.
3. **DHCP relay is transparent but must be present.** Every sub-interface that serves clients needs `ip helper-address`; omitting even one VLAN leaves those clients without leases.
4. **ISP return route is essential.** Without a static route on the ISP pointing back to the PAT address, replies never reach inside clients even though outbound flows succeed.
5. **Trunk hygiene prevents STP issues.** Restricting allowed VLANs on trunks and enabling PortFast/BPDU Guard on user ports eliminates unnecessary topology changes.

Copyright & Attribution

© Yusuf

All rights reserved.

Source & project files: github.com/engYSF

This report was produced as a final project submission for the Faculty of Computers & Information Technology, University of Tabuk.